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A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

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ABSTRACT Sleep-disordered breathing is common in the weeks after an ischemic stroke and predicts worse recovery, yet automatic sleep analysis reads only the electroencephalogram and reports only sleep stages. A clinical polysomnogram records far more, and in this population the disease is plainly visible in those cardiorespiratory channels: the median patient here has a scored respiratory event in 13% of epochs. We present MM-Net, a two-stream multi-task model that reads fourteen neural and cardiorespiratory channels of the polysomnogram and produces two clinical outputs from a single forward pass, the sleep stage of every 30-second epoch and a per-epoch respiratory-event label, with a direct cardiorespiratory path carrying breathing effort to the respiratory decision undiminished by the staging objective. On iSLEEPS, a stroke-specific polysomnography corpus (99 patients, 92,560 epochs), under patient-independent ten-fold cross-validation, MM-Net stages sleep at 0.739 accuracy (Cohen's κ 0.634) and detects respiratory events at 0.782 AUC with 2.76 M trainable parameters. We benchmark thirteen deep architectures retrained on this cohort and three applied zero-shot from healthy sleep; every retrained baseline lands between 0.613 and 0.720 accuracy, and those we ran ourselves share the proposed model's folds. Six are reimplemented here, each recovering its published accuracy to within 0.04, which places the limit in the cohort rather than in the implementations. Cross-validated on healthy sleep the same architecture reaches 0.856, losing 0.115 crossing into stroke against 0.140 to 0.148 for those baselines. Applied zero-shot to two independent corpora, the respiratory read-out could be scored only on ISRUC, which alone carries cardiorespiratory channels, reaching 0.780 AUC on its first night and 0.723 on its second. Code, features and checkpoints are released.

INDEX TERMS Cardiorespiratory signals, feature fusion, ischemic stroke, multi-task learning, multimodal learning, polysomnography, sleep apnea, sleep staging.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11]. On healthy sleepers these methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagreement [12], [13].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing shows: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We read the polysomnogram across both neural and cardiorespiratory physiology, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [14], released in 2026 and described by its authors as the first Asian and one of the largest stroke-specific sleep databases. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [15], [16], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 79 of 99 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [17], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark on the same folds. Our contributions are:

1) We introduce MM-Net, a two-stream multi-task model that stages sleep at 0.739 accuracy (κ 0.634) and, from the same forward pass, detects respiratory events at 0.782 AUC, with a direct cardiorespiratory path to the respiratory head so cardiorespiratory cues reach the decision undiminished by a fusion trained mostly for staging; removing that path costs 0.006 respiratory AUC and leaves staging unchanged (Section V). The ablation identifies respiratory effort, not oxygen saturation, as the channel the read-out depends on (Section VI-F).
2) We benchmark the model against sixteen baseline configurations, those we ran ourselves on the same patient-independent folds as the proposed model, covering convolutional, recurrent, attentional and fully convolutional families, and reimplement six published architectures (U-Time, TinySleepNet, SleepTransformer) so that every comparison runs through one pipeline (Table 4).
3) We establish that the accuracy ceiling on this cohort belongs to the patients and not to our implementations. Each reimplementation recovers its published Sleep-EDF accuracy to within 0.04, every retrained baseline lands between 0.613 and 0.720 here, and the proposed model loses 0.115 accuracy crossing from healthy sleep to stroke against 0.140 to 0.148 for those baselines (Section VI-M).
4) We show the respiratory read-out is physiologically grounded: removing the airflow channel from which the events were scored leaves detection at 0.778 AUC, because the decision rests on respiratory effort above all, together with oxygen desaturation, cardiac variability and cortical arousals, encoded redundantly enough that no single channel is indispensable (Section VI-H).
5) We attribute each output to the physiology it depends on, by intervention on the model rather than by correlation. A leave-one-out modality-ablation grid separates the two outputs: removing the cardiorespiratory stream lowers respiratory detection and leaves staging untouched (-0.128 AUC, Wilcoxon on ten fold-means, Holm $p = 0.018$), while no neural ablation moves respiratory AUC. That double dissociation is a falsifiable account of what each modality contributes (Table 6).

Together, these results reframe the problem on this cohort: the useful gains come from reading the whole sleeping body and attributing each output to the physiology that produces it; deeper single-channel staging does not help on this cohort. The remainder of the paper develops the model (Section V) and tests each claim in turn.

II. RELATED WORK

A. SINGLE-CHANNEL DEEP SLEEP STAGING

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; and XSleepNet fused raw and spectral views [6]. More recent backbones push this further: SleepTransformer adds interpretability and uncertainty to an attention model [18], L-SeqSleepNet models whole-night cycles [19], ZleepAnlystNet and FlexSleepTransformer explore separated training and flexible channel sets [20], [21], and sleep stagers built on the linear-time state-space backbone of Mamba [22] reach competitive accuracy [23]. These models set the accuracy on healthy corpora but depend on the cortical micro-events that stroke disturbs, which is why they transfer poorly to patients [17].

B. MULTI-CHANNEL AND FOUNDATION MODELS

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [24], and self-supervised EEG foundation models such Joint scoring of the two tasks is itself established. Huttunen et al. trained a single network to score sleep stages and respiratory events simultaneously across 877 clinical recordings, comparing signal combinations from pulse oximetry alone up to a full montage [25], and multitask staging-and-apnea models have followed [26]. Those studies work in suspected-apnea and elderly populations; what is untested is whether the arrangement survives a cohort whose neural signal is itself damaged, which is the question this paper asks. Foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8]–[11]. SleepFM extends this idea to several PSG modalities [27], while cross-scale foundation models and standardized benchmarks continue to appear [28]–[30], self-supervised pretraining is now systematically evaluated for label efficiency [31], transfer and domain adaptation improve robustness across cohorts [32], and recent surveys and clinical benchmarks catalogue the field [13], [33]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

C. DETECTION OF SLEEP-DISORDERED BREATHING

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is part of the scoring criterion for a hypopnea [1]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

D. SLEEP AND BREATHING AFTER STROKE

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [15], [16], [34]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [35]–[39]; quantitative EEG markers such as the brain symmetry index further track hemispheric injury [40]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population. Sleep staging on ischemic stroke itself is nascent, reported only as baselines on the iSLEEPS corpus [14], and no prior method addresses staging and respiratory detection jointly, on a stroke population; we make this comparison quantitative in Table 13.

III. CRITICAL GAPS AND LIMITATIONS OF PRIOR WORK

Three gaps run through the literature above, and each maps onto one of our contributions. *(i) Single-task, single-signal design.* Almost every staging model reads one or a few EEG channels and predicts only the stage, while apnea detectors read breathing signals and predict only events. Multimodal foundation models have begun to join them, SleepFM [27] staging sleep and classifying apnea severity from one representation on a general population, but per-epoch respiratory detection alongside staging has not been shown on a stroke cohort, leaving the clinician with two separate tools. Contribution 1 answers this with a single model that reads both physiologies and produces both outputs. *(ii) Evaluation on healthy sleepers.* Staging accuracy is reported almost entirely on healthy corpora such as Sleep-EDF [41], [42], where deep and foundation models reach a ceiling, while performance on focal brain injury is largely unmeasured. Contributions 2 and 3 benchmark strong deep architectures, and classical gradient-boosting pipelines [43]–[45] on a subacute-stroke cohort, show that the healthy-built networks collapse, and establish by reimplementation that the collapse is a property of the cohort. *(iii) Attribution and reliability.* Interpretability is usually post-hoc and unfalsifiable, and predictive uncertainty is seldom modeled [46]. Contributions 4 and 5 answer this with a physiologically grounded, non-circular respiratory read-out and an interventional modality-ablation that attributes each output to the channel whose physiology is closest to it.

TABLE 1. **Stage distribution and respiratory-event prevalence in the corpus we model ($N = 99$). Respiratory events occupy 16.0% of all epochs, and the per-patient burden ranges from 0 to 60.5%, with 79 of 99 patients above 5% (Fig. 1).**

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

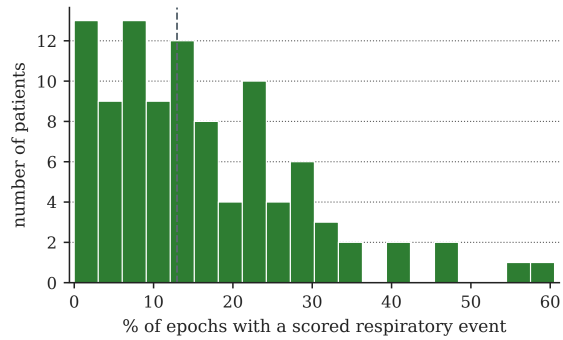

FIGURE 1. **Sleep-disordered-breathing burden across the cohort: patients by the fraction of epochs carrying a scored respiratory event, over the 99 with respiratory labels. The dashed line marks the median, 13%.**

IV. DATASET AND PREPROCESSING

iSLEEPS [14] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments, giving $N = 99$. Both tasks use all 99 patients: the twelve recordings that lack at least one cardiorespiratory channel are scored with those channels zero-filled rather than excluded, and we state which N applies to each reported number. The corpus totals 92,560 epochs. The stage distribution (Table 1) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

a: Channels we read, and channels we do not

A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO_2, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO_2 and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz for the engineered descriptors, and independently to 200 Hz for the pretrained encoder as described below; the cardiorespiratory channels go to a common 25 Hz grid; SpO_2, recorded natively at 4 Hz, is upsampled to this grid, so its effective resolution remains that of the 4 Hz oximeter. Signals are read in physical units, which matters because SpO_2 near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an event.

V. PROPOSED METHOD

A. PROBLEM FORMULATION

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eeg}} \in \mathbb{R}^{388}$, formed by concatenating the 188 engineered descriptors below with a 200-dimensional embedding from LaBraM, a frozen pretrained EEG encoder [8] (Section VI-N), and a raw cardiorespiratory tensor $\mathbf{x}_t^{\text{car}} \in \mathbb{R}^{7\times 750}$, the seven channels at 25 Hz. The 14 engineered cardiorespiratory descriptors defined below are used by the baseline configuration against which the learned encoder is measured.

The pretrained encoder is fed on its own terms rather than on ours. LaBraM was pretrained at 200 Hz and fixes its input length at 3000 samples, so the EEG is resampled from our 100 Hz working rate to 200 Hz before it reaches the encoder. Feeding the 100 Hz signal directly would have been silent but wrong: 3000 samples would then span 30 rather than 15 seconds and every rhythm would present at half its frequency, so a 12 Hz spindle would arrive as 6 Hz. At 200 Hz a 30 s epoch no longer fits the positional embedding in one piece, so each epoch is split into two 15 s sub-windows of 3000 samples, encoded separately, and mean-pooled; the two embeddings were checked to differ before caching, so the pooling carries information rather than averaging duplicates. The encoder also requires canonical 10–20 names and rejects mastoid references, so C4:M1, C3:M2, O2:M1 and O1:M2 are presented as C4, C3, O2 and O1, and the ocular and chin channels are dropped for having no canonical EEG name. Amplitudes are scaled to μV/100. The encoder therefore sees four EEG channels, strictly fewer signals than the engineered descriptors read. We learn a subject-independent map

$$f_\theta : \left(\mathbf{f}_{1:T}^{\text{eeg}}, \mathbf{x}_{1:T}^{\text{car}}\right) \longmapsto \left(\hat{y}_{1:T}^{\text{stg}}, \hat{y}_{1:T}^{\text{apn}}\right), \tag{1}$$

where $\hat{y}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{y}_t^{\text{apn}} \in [0,1]$ is the probability that epoch t contains a respiratory event (Fig. 2).

B. PHYSIOLOGICAL FEATURE REPRESENTATIONS

a: Neural features (188)

Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–13), sigma (11–16), and beta (16–30) Hz; the relative power in band b is

$$\mathrm{RP}_b = \frac{\int_{f\in b} P(f)\,df}{\int_{0.5}^{30} P(f)\,df}. \tag{2}$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [47], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope $e(t) = |\mathcal{H}\{x_\sigma\}(t)|$ from the Hilbert transform $\mathcal{H}$, with density counted above a per-epoch threshold; slow-wave amplitude, ocular-movement energy on the EOG, and tonic EMG level complete the set.

b: Cardiorespiratory features (14)

From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath. The correlation is signed, so synchronous and paradoxical motion sit at opposite ends of the range rather than being folded together,

$$\mathrm{asyn}_t = \mathrm{corr}(x_t^{\text{tho}}, x_t^{\text{abd}}). \tag{3}$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people. Appendix A lists all 188 neural descriptors with the parameters used to compute them.

c: The learned cardiorespiratory encoder

The evaluated model does not read the 14 descriptors. Its cardiorespiratory branch is a three-layer convolutional encoder over the raw 7×750 tensor, with wide kernels and aggressive downsampling because respiratory events are slow: at 25 Hz the first kernel of 25 samples spans one second, and a hypopnea unfolds over 10 to 30. Channel widths are 48, 96 and 96, each followed by batch normalisation and GELU, with max-pooling by four after the first two. Mean and max pooling over time are concatenated and projected to the same 64 dimensions the feature MLP produced, so the encoder is a drop-in replacement and every other part of the network is unchanged. Section VI-N reports what that substitution is worth, and it is the single largest architectural effect in this paper.

C. TWO-STREAM ENCODERS

The neural stream is a two-layer perceptron with LayerNorm; the cardiorespiratory stream is the convolutional encoder de-

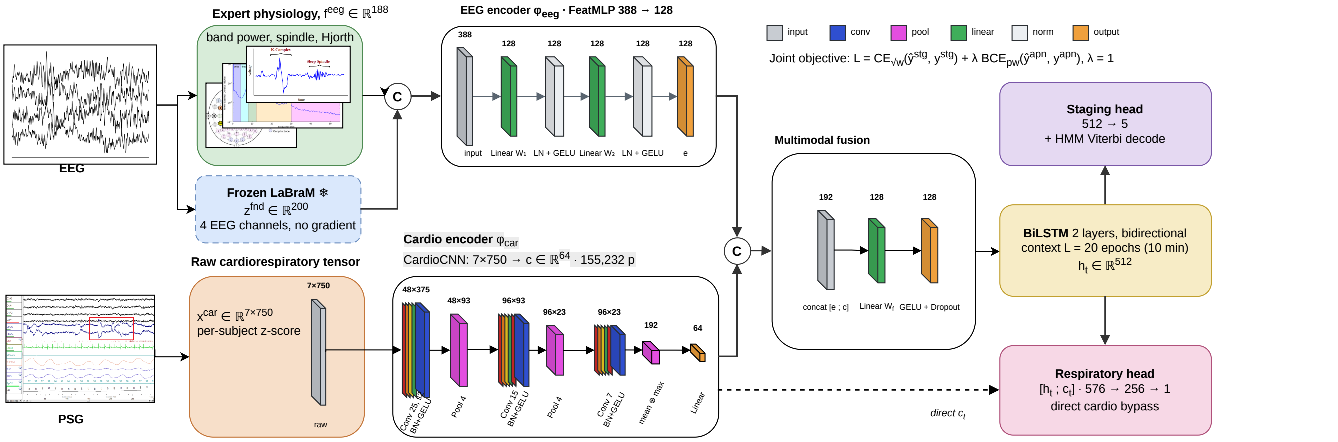

FIGURE 2. **The two-stream, multi-task model. Layer shapes above each block, operation below; block height encodes units for the multilayer perceptron and temporal length for the convolutional stack. Dashed outlines mark frozen parameters. The junction marked C is the concatenation used for every reported result, and the fusion panel is the single linear layer that follows it.**

scribed above, which emits a vector of the same width the perceptron it replaced produced,

$$e_t = \mathrm{GELU}\big(\mathrm{LN}(W_2\,\mathrm{GELU}(\mathrm{LN}(W_1\mathbf{f}_t^{\mathrm{eeg}})))\big) \in \mathbb{R}^{128}, \quad (4)$$
$$c_t = \mathrm{CardioCNN}\big(\mathbf{x}_t^{\mathrm{car}}\big) \in \mathbb{R}^{64}. \quad (5)$$

LayerNorm rather than BatchNorm is used because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because the branch compresses a 7×750 tensor into the same 64 dimensions the feature encoder it replaced produced, which keeps the substitution the only variable.

D. FUSION

The two embeddings are concatenated and projected back to the stream width,

$$z_t = \mathrm{GELU}\big(W_f\,[\,e_t\,;\,c_t\,]\big) \in \mathbb{R}^D, \qquad D = 128. \quad (6)$$

This is deliberately the simplest fusion available, and it is what every number in this paper was produced with. We arrived at it by evaluating the obvious alternative first: the two embeddings as tokens carrying learned modality-type embeddings, refined by four-head scaled dot-product attention so each modality can read the other, then projected. On the same ten folds, in the feature-only configuration the comparison was run in, attention was indistinguishable from concatenation on staging (+0.005 accuracy, $p = 0.49$) and slightly worse on the respiratory head (-0.014 AUC, $p = 0.06$). Since the elaborate fusion buys nothing measurable here, we report and release the simple one, and the gain from the second stream should be read as coming from the added modality.

E. TEMPORAL DECODER AND TWO HEADS

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \quad (7)$$

and the forward and backward states of the 256-unit layers are concatenated into $h_t \in \mathbb{R}^{512}$. The staging head is a linear map $\hat{y}_t^{\mathrm{stg}} = \mathrm{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t,

$$\hat{y}_t^{\mathrm{apn}} = \sigma\big(W_2\,\mathrm{GELU}(W_1[h_t; c_t])\big). \quad (8)$$

The connection exists so that the respiratory decision does not have to survive a fusion trained mostly for staging, which would weaken the mechanical signature of an obstructed breath along the way. Section VI-F identifies the channel that signature lives in, and it is respiratory effort that names the event. We isolate the connection by removing it and holding everything else fixed, on the same folds as every other figure here. Respiratory AUC falls from 0.782 to 0.776 (Wilcoxon on ten fold-means, $p = 0.049$) and average precision from 0.419 to 0.411 ($p = 0.28$), while staging is untouched at 0.739 accuracy ($p = 0.92$) and κ 0.634 ($p = 0.92$). The gain is small, about a sixth of a fold standard deviation, and the p-value is marginal enough that we would not rest an argument on it alone. What makes it worth reporting is where it lands rather than how large it is: the connection feeds the respiratory head and nothing else, and the respiratory head is the only output that moves.

The trained model holds 2.76 million parameters. The pretrained encoder that supplies the 200-dimensional neural embedding is run once offline and is not trained here, so it adds no trainable parameters to this count.

F. TRAINING AND DECODING

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda\,\mathcal{L}^{\text{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\text{stg}} = -\sum_t \sum_k w_k\, y^{\text{stg}}_{t,k} \log \hat{y}^{\text{stg}}_{t,k}, \tag{9}$$

$$\mathcal{L}^{\text{apn}} = -\sum_t \left[\pi\, a_t \log \hat{y}^{\text{apn}}_t + (1-a_t)\log(1-\hat{y}^{\text{apn}}_t)\right]. \tag{10}$$

The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate. At inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transition matrix A^{H} and prior are counted from the training labels [48],

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i) + \log A^{\text{H}}_{ij}\right] + \log \hat{y}^{\text{stg}}_t(j), \tag{11}$$

recovered by back-tracking, which restores the natural persistence of sleep. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

VI. EXPERIMENTS

A. PROTOCOL AND METRICS

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K}\sum_k \text{F1}_k$, and Cohen's κ. The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

Every number we report is a mean over the ten folds with its standard deviation, and the folds are fixed across all configurations so that comparisons are paired. Fixing them buys paired comparisons at a price we should state plainly: the configuration reported here was chosen by comparing forty-nine candidates on these same ten folds, so the folds that selected the model are the folds that score it, and the in-domain figures are not free of selection optimism. Two things bound how much that can matter. The ten best candidates span 0.004 accuracy against a fold-to-fold standard deviation of 0.019, so the margin available to selection is about a fifth of the uncertainty we already report; and the external corpora in Section VI-P took no part in selection, so the zero-shot figures there are unaffected by it. A nested selection loop would remove the concern rather than bound it, and we did not run one. The proposed model, its ablations and the six reimplemented literature models all use the full ten; the earlier retrained baselines were run at five, which Table 4 states per row. Because a single training seed can flatter a result, we retrained the headline configuration under three seeds on the same folds. Staging accuracy varies by 0.005 across seeds against 0.020 across folds, and respiratory AUC by 0.001 against 0.038: fold-to-fold spread is four times the seed spread for staging and thirty times it for respiratory detection. The uncertainty in these results is therefore dominated by which patients fall in which fold, which is a property of a cohort this size. Every headline figure in this paper is the mean over all thirty fold-values, ten folds by three seeds. Paired tests, however, are computed on the ten fold-means: the three seeds within a fold train and test on the same patients, so they are three re-rolls of one experiment and not three independent observations, and treating them as thirty would triple the effective sample and shrink every p. One consequence is worth stating plainly. With ten paired folds the smallest two-sided Wilcoxon p is $2/2^{10} = 0.002$, reached when every fold moves the same way, so after Holm correction across a family of nine the floor is 0.018. Several effects below sit at that floor and differ in size by more than an order of magnitude; the test cannot separate them, and the effect sizes are reported beside every p for that reason. The three seed means for respiratory AUC are 0.781, 0.783 and 0.781, so 0.782 is stable to the third decimal under reseeding.

TABLE 2. Training configuration, shared across all folds, re-implemented baselines, and ablations.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 (6 GB)
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Staging class weights	$\sqrt{\ }$ inverse-frequency
Respiratory pos. weight	$\pi = n_-/n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seeds	42, 1, 7

B. IMPLEMENTATION AND HYPERPARAMETERS

The model is implemented in PyTorch and trained on a single NVIDIA RTX 2060 GPU (6 GB) under Windows. Every run is seeded, so each is individually reproducible; headline figures average three seeds (42, 1, 7) over the ten folds, giving the thirty fold-values used throughout. Where a result is derived from stored predictions rather than retrained, it comes from the seed-42 run, and the text says so at that point. Table 2 lists the training configuration, which is identical across all folds, all baselines we re-implement, and every ablation, so that reported differences come from the model.

TABLE 3. Reimplementation check for the models marked † in Table 4: ten-fold subject-independent cross-validation on the Sleep-EDF Cassette subjects 0–20, 40 recordings, single Fpz-Cz derivation. ≈ marks an accuracy inferred from a paper's macro-F1. Each published figure follows its own paper's Sleep-EDF protocol, which differs in subset and split.

Model	Published	Ours	Δ
U-Time [7]	≈ 0.79	0.791 ± 0.062	$+0.001$
TinySleepNet [4]	0.854	0.824 ± 0.045	-0.030
SleepTransformer [18]	0.814	0.823 ± 0.033	$+0.009$
1D-ResNet-SE-LSTM [49]	0.864	0.832 ± 0.036	-0.032
Micro SleepNet [50]	0.828	0.800 ± 0.041	-0.028
AT-BiLSTM [51]	0.838	0.835 ± 0.046	-0.003

C. COMPARED METHODS

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN–BiLSTM), six further published architectures we reimplemented for this comparison (U-Time, a fully convolutional encoder–decoder; TinySleepNet, a compact convolutional–recurrent model; and SleepTransformer, a purely attentional one), a network transferred from Sleep-EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [14]. All numbers are patient-independent, ten-fold except for the earlier retrained baselines at five, at the hidden-Markov operating point where a model provides one.

D. STAGING BENCHMARK AND COMPARISON

The row marked ‡ is quoted from the corpus paper rather than run here. It uses that paper's protocol, an $80/10/10$ patient-wise split over $N = 100$ that includes the duplicated pair we exclude (Section IV), and its own two-channel input, and that paper's repository releases preprocessing only, so it can be neither reproduced nor read as a same-protocol comparison. Each reimplementation is checked against its published Sleep-EDF accuracy in Table 3; those published figures follow each paper's own protocol, which differs in subset and split, so they are consistency checks rather than exact reproductions. Our own split uses one subject more than the conventional Sleep-EDF-20 set, and one of the published figures is measured on SHHS rather than Sleep-EDF, so those two checks are weaker than the rest. Table 4 is organised in three blocks: architectures retrained on this cohort, architectures trained on healthy sleep and applied to it zero-shot, and the proposed model, with a final row cross-validating the proposed architecture on healthy sleep as a control on the cohort argument; its macro-F1 and κ are healthy-sleep figures and so are left out of the stroke columns, and quoted in the text instead. Dispersion is reported wherever it is defined: five folds for the earlier retrained baselines; thirty fold-values, ten folds by three seeds, for AttnSleep, the three reimplemented literature models and the proposed rows. CNN-ResNet18 is quoted from the corpus paper rather than re-run here, so its protocol is that paper's and it carries no interval. That paper's LSTM appears only as our reimplementation of it on the folds used throughout, because its published figure was never measured on these folds. In the zero-shot block the interval is across pretraining seeds and not folds, because that evaluation is a single pass of a finished checkpoint over all 99 patients. The two best healthy scores, 0.856 and 0.855, are separated by less than either interval. Sleep-EDF carries no cardiorespiratory channels, so every healthy-sleep figure exercises the neural branch alone and the comparison row there is MM-Net (neural only).

Table 4 shows a clear pattern: every deep network, whether trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal, stays between 0.61 and 0.72 accuracy, 12 to 20 points below the same architectures' published accuracy on healthy sleepers. The feature-based model reaches 0.739, above every deep architecture we retrained here, which matches the concurrent finding that depth alone does not help on this cohort [17]: with under a hundred patients a convolutional network cannot rediscover what decades of sleep science already encode in the features. The corpus paper's own recurrent baseline, reimplemented here and run on the same folds, reaches 0.720 ± 0.032 over thirty fold-values. Staging accuracy is nonetheless capped: nothing we have run on this corpus exceeds 0.75. Section VI-L shows why: the staging learning curve has flattened, so the ceiling is a property of this cohort rather than of any one model. We therefore do not position the model as a staging leader. What sets it apart is a second output: it is the only model that also detects respiratory events. A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night's cycles, and the stage-probability ribbon exposes where the model is confident and where it hesitates.

The margin over the retrained baselines is tested. The folds are fixed across every model in Table 4, so each baseline is scored on exactly the patients the proposed model was and the comparison is paired. That matters here: fold-to-fold variation is large and it is *shared*, because the difficulty lies in the patients (AttnSleep spans 0.627 to 0.755 across folds), so reading two overlapping ±SD intervals charges the cohort's own variability against the difference and hides it. Paired on the ten fold-means, with Holm correction across the eight baselines we ran ourselves, the proposed model is ahead of every one of them on accuracy, by $+0.019$ to $+0.093$, and wins between eight and ten of the ten folds in each comparison. The margin is statistically separable for five: Micro SleepNet, SleepTransformer, U-Time, AT-BiLSTM and TinySleepNet (Holm p from 0.016 to 0.039). For the three closest it is not. AttnSleep ($+0.023$, Holm $p = 0.17$), our reimplementation of the corpus LSTM ($+0.019$, $p = 0.17$) and 1D-ResNet-SE-LSTM ($+0.032$, $p = 0.15$) sit inside the noise that ten folds can resolve. We report that as it stands, and it is the cohort-ceiling argument arriving by another route: three architectures spanning 2021 to 2023 land close enough to the proposed model that this cohort cannot tell them apart. The accuracy margin is the narrowest of the three and the one most exposed to the cohort's noise; macro-F1 and κ, which do not let a

TABLE 4. Sleep staging on iSLEEPS, patient-independent, against every architecture we evaluated. Cells are mean ± SD across folds where per-fold values were retained. † marks a model we reimplemented (Table 3), ‡ one quoted from the corpus paper on its own protocol, "feat" engineered features. Bold marks the best value among rows run under our protocol.

Model	Input	Params ↓	Healthy	iSLEEPS (stroke)		
			Acc	Acc ↑	Macro-F1 ↑	κ ↑
Deep networks, retrained on iSLEEPS						
LSTM, our reimplementation†	1 EEG + EOG	1.16 M	—	0.720 ± 0.032	0.626 ± 0.028	0.599 ± 0.043
CNN-ResNet18 [17]‡	1 EEG + EOG	~11 M	—	0.617	0.544	0.480
DeepSleepNet [2]	1 EEG	~21 M	—	0.615 ± 0.026	0.567 ± 0.031	0.483 ± 0.038
U-Time [7]†	1 EEG	1.07 M	0.791 ± 0.062	0.646 ± 0.064	0.524 ± 0.057	0.506 ± 0.082
AttnSleep [5]	1 EEG	~0.5 M	—	0.716 ± 0.042	0.600 ± 0.043	0.593 ± 0.058
TinySleepNet [4]†	1 EEG	1.45 M	0.824 ± 0.045	0.676 ± 0.050	0.587 ± 0.034	0.548 ± 0.062
SleepTransformer [18]†	1 EEG	2.79 M	0.823 ± 0.033	0.683 ± 0.026	0.602 ± 0.030	0.555 ± 0.032
1D-ResNet-SE-LSTM [49]†	1 EEG	0.65 M	0.832 ± 0.036	0.708 ± 0.038	0.631 ± 0.033	0.592 ± 0.050
Micro SleepNet [50]†	1 EEG	0.03 M	0.800 ± 0.041	0.647 ± 0.049	0.558 ± 0.028	0.510 ± 0.055
AT-BiLSTM [51]†	1 EEG	0.92 M	0.835 ± 0.046	0.692 ± 0.040	0.612 ± 0.030	0.570 ± 0.051
CNN + BiLSTM	4 EEG	~0.6 M	—	0.613 ± 0.040	0.565 ± 0.034	0.485 ± 0.048
Sleep-EDF transfer	4 EEG	~0.6 M	—	0.623 ± 0.020	0.582	0.488
Raw multimodal CNN	14 ch	0.95 M	—	0.654 ± 0.033	0.610 ± 0.022	0.544 ± 0.037
Trained on healthy sleep, applied zero-shot						
CNN	4 EEG	0.49 M	0.806 ± 0.001	0.307 ± 0.016	0.223 ± 0.013	0.102 ± 0.010
CNN + BiLSTM (CRNN)	4 EEG	1.15 M	0.828 ± 0.005	0.312 ± 0.012	0.253 ± 0.006	0.127 ± 0.022
StagingSeqNet	4 EEG	1.08 M	0.855 ± 0.001	0.306 ± 0.009	0.230 ± 0.014	0.091 ± 0.015
Proposed feature-based multi-task model						
MM-Net (neural only)	7 ch feat	2.62 M	—	0.741 ± 0.017	0.668 ± 0.020	$\mathbf{0.635 \pm 0.022}$
MM-Net (neural + cardio)	7 ch feat + raw	2.76 M	—	0.739 ± 0.020	$\mathbf{0.670 \pm 0.020}$	0.634 ± 0.024
Same model, healthy sleep						
MM-Net (neural only)	2 EEG feat	2.62 M	$\mathbf{0.856 \pm 0.027}$	—	—	—

model coast on the two majority stages, separate the models by two to three times as much.

E. PER-CLASS STAGING PERFORMANCE

Table 5 breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and REM are recovered well, N3 is moderate, and N1 is the hard stage, because it is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the proposed model leads on the stable stages. The row-normalised confusion matrix (Fig. 4) shows where the errors land: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2. Comparing the neural-only and full rows isolates what the cardiorespiratory stream contributes to staging, and the answer is narrow: N1 improves by 0.019 (Wilcoxon $p = 0.11$ on ten fold-means, not significant) and no stage changes significantly ($p \geq 0.15$; REM shifts by 0.01 in Table 5, well inside that). That the gain lands on N1 is consistent with its physiology. It is the transitional stage, the hardest to score, and the one most often coincident with a respiratory event.

F. MODALITY-ABLATION GRID

This ablation is also our attribution method. Because the model is feature-based and never sees a raw spectrogram or a raw EEG trace, a gradient heat-map or Grad-CAM would have no faithful spatial support on the neural side, which is where a staging attribution would be read; we therefore attribute each output by *interventional* ablation, deleting a physiological channel and measuring the change. This is a falsifiable test of what each modality contributes. The proposed model stages sleep at 0.739 accuracy (κ 0.634) and detects respiratory events at 0.782 AUC (0.419 average precision). To test the paper's central claim, that each modality carries the task its physiology is closest to, we remove each modality in turn and keep the rest, on the same folds, with three seeds giving thirty fold-values per row (Table 6). The separation between streams is clean, but within the cardiorespiratory stream the pattern is not the one the engineered features gave.

Removing the whole cardiorespiratory stream drops respiratory AUC from 0.782 to 0.654, the largest single effect in this study, and leaves staging untouched (0.741, n.s.). Removing the neural stream does the reverse: staging falls from 0.739 to 0.680 while respiratory detection holds. No neural ablation moves respiratory AUC and no cardiorespiratory ablation moves staging, so each stream serves the outcome its physiology is closest to.

TABLE 5. Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point. The neural-only and full rows are means over the same thirty fold-values. Best per column in bold.

Model	W ↑	N1 ↑	N2 ↑	N3 ↑	R ↑
Raw multimodal CNN	0.74	**0.37**	0.73	0.56	0.67
MM-Net (neural only)	**0.79**	0.28	0.80	0.71	**0.76**
MM-Net (neural + cardio)	0.79	0.30	**0.80**	**0.71**	0.75

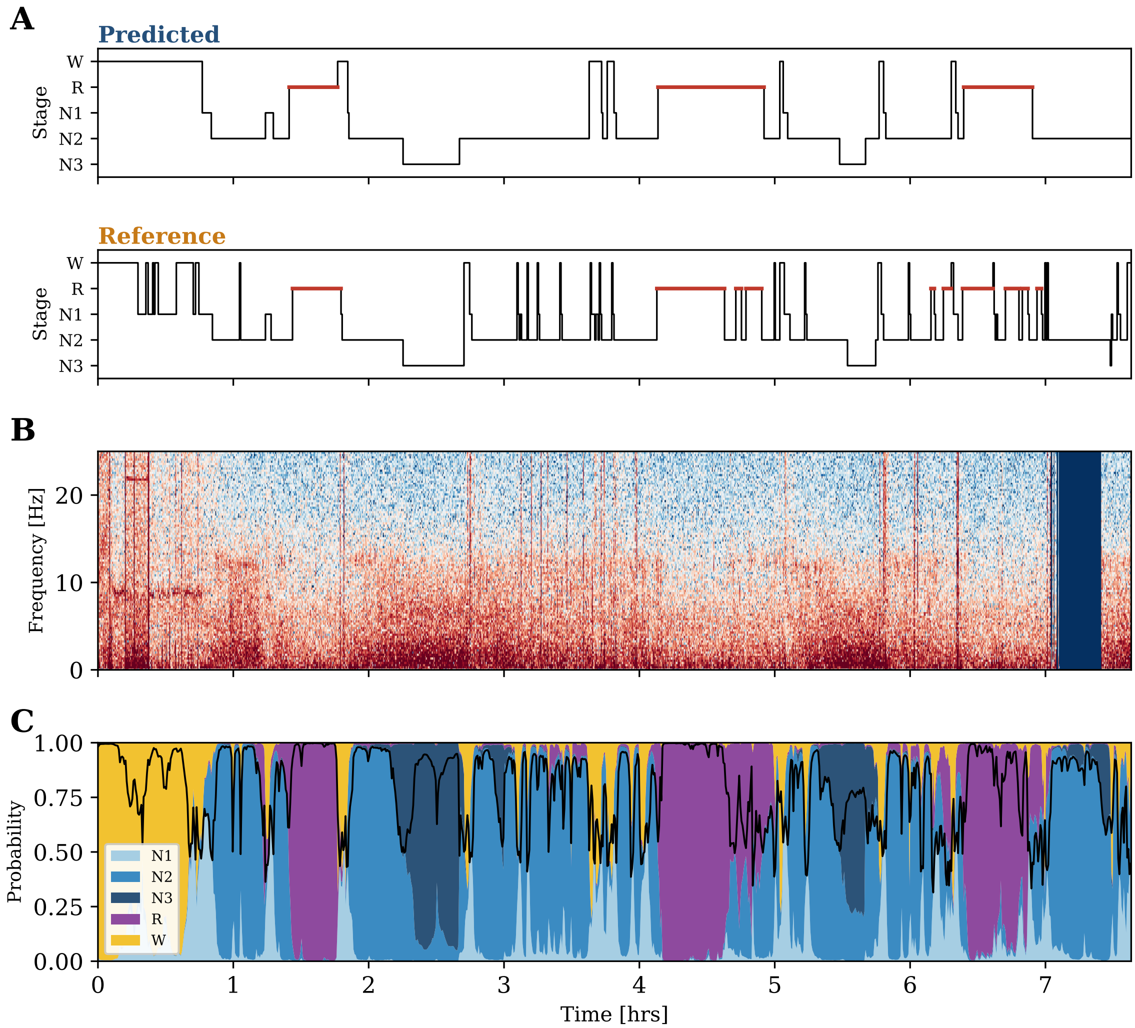

FIGURE 3. A representative held-out night (patient SN90, staging accuracy 0.80). (A) Predicted and reference hypnograms, REM highlighted. (B) EEG spectrogram, C4 derivation, 0–25 Hz. (C) Per-epoch stage-probability ribbon; the black line is the maximum posterior.

Within the cardiorespiratory stream, however, only respiratory effort is individually necessary (-0.038 AUC, Holm $p = 0.018$). Removing SpO_2, pulse/HRV, ECG or airflow individually costs at most 0.005 AUC and none survives correction; SpO_2, which the engineered-feature configuration was built around, is here among the least necessary channels (-0.002, $p = 1.00$). The five individual removals sum to -0.046 AUC against -0.128 for removing the stream as a whole, a factor of 2.8. The learned encoder therefore distributes respiratory information redundantly across channels: each is individually dispensable because the others overlap it, while the ensemble is essential. That redundancy is clinically favourable, since twelve of the ninety-nine recordings are missing at least one cardiorespiratory channel, and it is physiologically coherent: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event as it happens, whereas desaturation is a lagged consequence of it.

G. RECORDINGS WITH AN INCOMPLETE MONTAGE

Those twelve recordings are scored with the absent channels zero-filled rather than being excluded, which raises a fair question: does the network read missingness instead of physiology? It does not appear to. Removing all twelve from the pooled respiratory evaluation moves AUC from 0.778 to 0.775 and moves average precision the other way, from 0.419 to 0.439, so the respiratory result does not rest on them. Predicted per-patient burden is independent of whether a montage was complete ($p = 0.91$), so the shortcut is not available even in principle.

Staging is lower on those twelve, 0.670 against 0.743 ($p = 0.017$), and that needs a control rather than an assurance. The neural-only configuration receives no cardiorespiratory channel at all and so has nothing zero-filled, yet it shows the same gap in the same direction, 0.689 against 0.744 ($p = 0.040$). Most of the difference therefore belongs to the recordings, which are harder for a model that cannot see the missing channels either. Stated precisely rather than

TABLE 6. **Modality-ablation grid (leave-one-out), ten-fold patient-independent, three seeds. Each row removes one modality and keeps the rest; cells are means over thirty fold-values. p is Wilcoxon on the ten fold-means, Holm-corrected across the nine ablations within each metric and bold where it clears 0.05; the floor at this sample size is 0.018.**

Modality removed	Staging acc ↑	Holm p	Respiratory AUC ↑	Holm p
none (full model)	0.739 ± 0.020	—	0.782 ± 0.038	—
EEG (neural)	0.680	**0.018**	0.774	0.92
EOG (ocular)	0.731	0.08	0.777	1.00
EMG (muscle)	0.735	1.00	0.778	1.00
SpO_2	0.742	1.00	0.780	1.00
respiratory effort	0.741	1.00	0.744	**0.018**
pulse / HRV	0.741	1.00	0.785	1.00
ECG	0.737	1.00	0.777	1.00
airflow	0.740	1.00	0.778	1.00
all cardiorespiratory	0.741	1.00	0.654	**0.018**

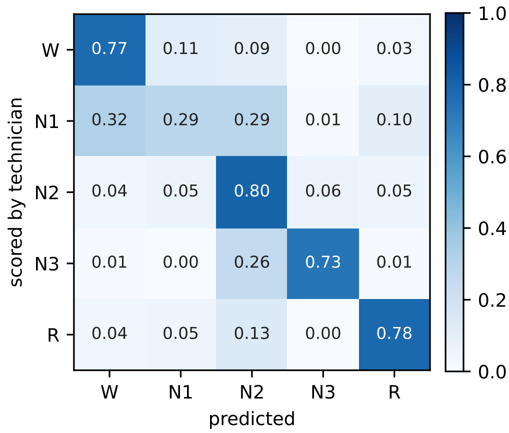

FIGURE 4. **Row-normalised confusion matrix for the proposed model (pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.**

generously: on those twelve the full model sits 0.019 below neural-only, which at $n = 12$ is not distinguishable from noise (Wilcoxon $p = 0.18$), so a small cost cannot be excluded, only bounded. On the eighty-seven complete recordings the two agree to within 0.002. Attention and plain concatenation stage and detect alike, so the gain comes from the added modality, and the fusion mechanism has little effect.

H. PHYSIOLOGICAL VALIDITY OF THE RESPIRATORY READ-OUT

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial reason. Removing the airflow features leaves detection essentially unchanged, at 0.778 AUC (Table 6), within one fold standard deviation of the full 0.782. The model therefore does not lean on the labeling signal. It reads above all the straining of the chest and abdomen (the only channel whose removal costs significantly), together with the fall in blood oxygen, the beat-to-beat change in the heart, and the cortical arousal that ends the event, none of which is individually indispensable because the learned encoder encodes them redundantly. This is the useful regime, because it detects the bodily consequences of disordered breathing, not a repeat of a measurement a technician has already made. It is also the physiologically expected one: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event while it is happening, whereas desaturation follows it by several breaths.

I. RESPIRATORY BASELINES AND PER-EVENT-TYPE DETECTION

The respiratory task needs real competitors (Table 8), on the same folds. The classical baselines read the 14 engineered cardiorespiratory descriptors; the proposed model reads the raw signal through its learned encoder, and its own score on the 14 descriptors is 0.698 (Table 10), which is the like-for-like comparison. A clinical desaturation rule reaches 0.596 AUC, logistic regression 0.582, and gradient boosting 0.670; a random classifier scores 0.157 average precision, the event prevalence. The proposed model reaches 0.782 AUC and 0.419 average precision, above gradient boosting but by a modest margin, so the contribution is the joint single pass and the physiological attribution above; we do not claim state-of-the-art detection accuracy. Detection also separates by event type: the model scores 0.748 AUC on hypopneas (the mild, dominant 81% of events), 0.898 on obstructive apneas, and 0.935 on central apneas. It is strongest on the most severe events, and the overall AUC is held down by the abundant hypopneas.

Scoring at the event level instead of the epoch level worsens the picture (Table 7). No threshold is comfortable: the one that finds most events flags two thirds of the night, and the one that flags least finds a quarter of them. More tellingly, the model recovers the wrong *number* of events, and its per-patient event count agrees with the clinical index far more weakly than its aggregated epoch probability does ($\rho = 0.142$ against $\rho = 0.491$). Consecutive events separated by a few epochs merge into one flagged run, so an epoch-level detector systematically under-counts. This is why we report a burden, and why event-level scoring is the natural next step instead of a formality.

Figure 5 shows the operating characteristics behind these numbers, computed from the pooled held-out predictions of all ten folds. The precision–recall panel is the more informa-

TABLE 7. Event-level detection, pooled over the ten patient-independent folds. Consecutive flagged epochs are grouped into events within each patient. The flagged fraction is the share of all epochs the threshold marks.

Threshold	Event sens. ↑	False alarms/h ↓	Epochs flagged
0.2	0.857	2.22	68%
0.3	0.785	2.35	57%
0.4	0.683	2.43	45%
0.5	0.549	2.61	33%
0.6	0.397	2.21	22%
0.7	0.270	1.41	13%
Per-patient agreement with the clinical index			
Aggregated epoch probability (burden)			$\rho = 0.491$
Counted events per hour			$\rho = 0.142$
Predicted vs scored event rate			4.5 vs 11.1

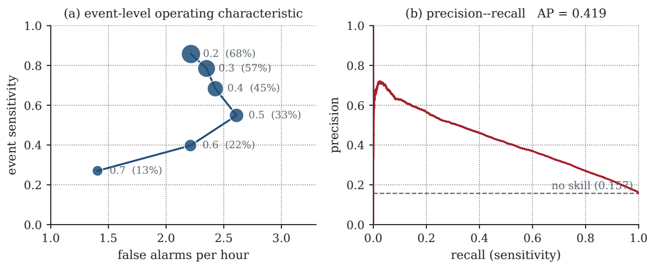

FIGURE 5. Respiratory-event detection, pooled over the ten patient-independent test folds. (a) Event-level operating characteristic. Each point is a decision threshold, labelled and sized by the fraction of all epochs it flags. (b) Precision–recall; the dashed line is the 0.157 event prevalence, the no-skill floor.

TABLE 8. Respiratory-event detection, ten-fold patient-independent. *(a)* Detectors on the same folds; the classical baselines read the 14 descriptors, MM-Net the raw signal. Chance average precision is the 0.157 prevalence. *(b)* AUC by event type, 96 annotated patients. Best per column in bold.

(a) Detection method

Method	AUC ↑	AP ↑
Desaturation rule	0.596	0.214
Logistic regression	0.582	0.193
Gradient boosting	0.670	0.290
MM-Net (proposed)	**0.782**	**0.419**

(b) MM-Net AUC by event type

Event type	Positive epochs	AUC ↑
Hypopnea	11,605	0.748
Obstructive	1,661	0.898
Central	831	**0.935**

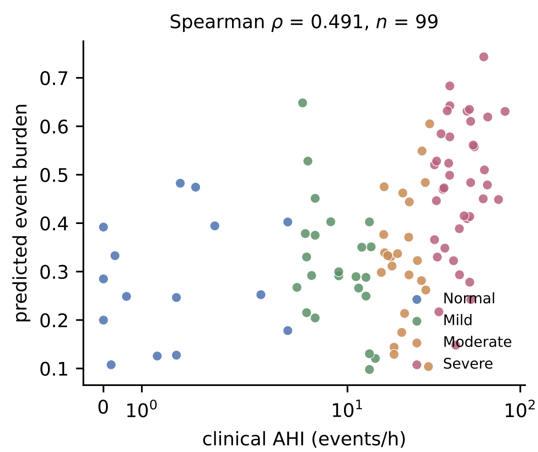

FIGURE 6. Predicted per-patient event burden (mean respiratory probability) against the clinical apnea-hypopnea index, coloured by AASM severity band. The positive association (Spearman $\rho = 0.491$, $n = 99$) shows the model's output tracks a clinician's summary measure.

tive of the two: respiratory events occupy 16% of epochs, and under that imbalance a ROC curve flatters any detector. Read at fixed sensitivity, the trade-off is explicit: at 0.80 sensitivity the model holds 0.47 specificity and 0.22 precision, which is a research-grade burden estimate, not a scoring-grade detector, and we describe it as such. We also note that pooling every fold's epochs into a single curve is not the same estimator as averaging the ten per-fold scores: pooling weights folds by epoch count and mixes their score scales, giving a pooled AUC of 0.777 against the cross-validated 0.782 ± 0.038 reported in Table 8. Both are stated so the difference is visible; at 0.005 it is an eighth of one fold standard deviation.

J. CLINICAL VALIDATION AGAINST AHI AND SEVERITY

Aggregating the per-epoch respiratory probability into a per-patient burden and correlating it with the clinical apnea-hypopnea index (AHI) from the metadata gives a significant positive association (Spearman $\rho = 0.491$, $p < 0.001$, $n = 99$, the twelve recordings with an incomplete cardiorespiratory montage scored with those channels zero-filled; Fig. 6), so the model's output tracks a clinician's summary measure beyond an epoch-level score. Staging, in turn, degrades as sleep-disordered breathing worsens: accuracy falls from 0.769 in patients with a normal AHI to 0.719 in the severe group (Table 9). The pathology that motivates the second output also makes the first harder, which is the clinical reality of this cohort.

The association is real but modest, and it is worth saying why it is so. Three mechanisms plausibly cap it, and each is a property of the formulation rather than of the model. First, mean per-epoch probability is a poor estimator of an hourly event rate: a 30-second epoch holding one hypopnea and one holding three both contribute a single positive, so the aggregate compresses exactly the variation the index is built from. Second, the clinical index is normalised by total sleep time, whereas our burden averages over every scored epoch including wake, and the two denominators diverge most in the patients with the most fragmented sleep, which in this cohort is the severe group. Third, the AHI itself carries substantial night-to-night variability, so a single-night reference sets a ceiling on any correlation with it. We therefore read $\rho = 0.491$ as evidence that the read-out tracks severity at the group level, without claiming that it estimates an index.

TABLE 9. Sleep-staging accuracy by sleep-disordered-breathing severity (AASM AHI bands), ten-fold patient-independent; mean ± SD across patients ($N = 99$). Staging degrades monotonically as severity rises.

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	15	0.769 ± 0.049
Mild	5–15	23	0.747 ± 0.069
Moderate	15–30	23	0.722 ± 0.088
Severe	≥ 30	38	0.719 ± 0.100

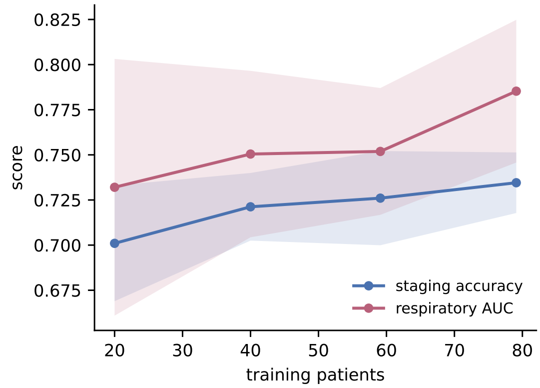

FIGURE 7. Learning curve over training-patient fraction, ten-fold patient-independent, error bars one standard deviation across folds. Staging (blue) flattens by 57 patients; respiratory detection is still climbing at 76. Bands are ± one standard deviation across folds.

K. PERMUTATION IMPORTANCE: A SECOND, INDEPENDENT ATTRIBUTION

The ablation grid establishes what the model *needs*: remove a modality, retrain, and observe which head suffers. A complementary question is what a *trained* model actually uses. We therefore permute each modality's features across epochs at inference time, so the channel is still present but no longer aligned to the epoch it describes, and measure the drop, averaged over three shuffles and the ten fold models.

The two methods can disagree, because a model may need a channel during training yet weight it lightly afterwards. On staging they do not: permuting the neural features costs 0.284 staging accuracy against 0.030 for the ocular and 0.006 for the muscle channel, the same ordering the ablation gives, and the rank correlation between the two attributions is $\rho = 0.833$ ($p = 0.010$).

On the respiratory head the agreement is narrower, and we state it precisely instead of pooling the two into a single coefficient. Both methods place respiratory effort first by a wide margin: 0.038 AUC under retraining, 0.071 under permutation, in each case well clear of the runner-up. The remaining four cardiorespiratory channels, however, have retraining effects that are not distinguishable from zero, so their rank order carries no signal there and the correlation over the five is correspondingly uninformative ($\rho = 0.381, p = 0.35$).

Permutation separates them where retraining cannot. Each fold records its own baseline AUC and its AUC after shuffling, so the drop is a within-fold quantity and can be tested directly. Every channel's drop is distinguishable from zero (one-sample on ten per-fold drops, Holm across the eight channels): effort 0.071 ($p < 10^{-4}$), airflow 0.027 ($p = 0.0005$), SpO_2 0.016 ($p = 0.0002$), pulse/HRV 0.012 ($p = 0.004$), and the remainder below 0.006. Significance here is not importance. The magnitudes span a factor of 32, and what the test establishes is that the *ordering* is real. The ordering is the result worth having: the three channels the model leans on most, effort, airflow and SpO_2, are exactly the three the AASM definition of a respiratory event is written in, and they are one to two orders above the cardiac and muscle channels.

The two experiments answer different questions, so the divergence is informative. Retraining without a channel measures whether it is *irreplaceable*, and the network compensates by re-learning the signal from its correlates; permutation holds the fitted model still and measures what it *uses*. SpO_2 is the clearest case: retraining without it costs 0.002 AUC, shuffling it under the fitted model costs 0.016. A channel can be heavily used and still be replaceable. The claim the comparison supports is therefore the one that matters physiologically: two attributions sharing no machinery, one applied to retrained models and one to fixed ones, independently identify effort as the dominant respiratory cue.

L. DATA EFFICIENCY AND THE COHORT CEILING

Staging on this corpus saturates near 0.72–0.75 for every model family tried here and elsewhere. Whether that reflects the architecture or the cohort is testable: retrain the identical model on a shrinking fraction of the *training patients*, always evaluating on the full held-out fold (Fig. 7). Subsampling is by patient, because epochs within a patient are heavily correlated.

The two heads answer differently. Staging accuracy rises $0.701 \rightarrow 0.721 \rightarrow 0.726 \rightarrow 0.735$ as training patients go from 20 to 79, with marginal gains of $+0.020$, $+0.005$ and $+0.009$, all under half a fold standard deviation and flattening. Respiratory AUC over the same range rises $0.732 \rightarrow 0.750 \rightarrow 0.752 \rightarrow 0.785$, and its *largest* increment is the final one, $+0.033$, close to a full fold standard deviation, and steeper than any earlier step. This curve is a single seed on a fixed held-out set, so its endpoint differs slightly from the three-seed headline figures (0.739 and 0.782); the shape is what this curve is here to show. Staging is at or near this cohort's ceiling; respiratory detection is not, and is still accelerating. We draw the narrower conclusion the data support: more patients of this kind would improve breathing detection and would not appreciably improve staging, so the staging ceiling is a property of the labels and the injured brain. The final model also exceeds the feature-only configuration at the full training-set size, so its advantage is not an artefact of the full sample.

M. HEALTHY-TRAINED MODELS APPLIED TO THE INJURED BRAIN

Table 4 retrains published architectures *on* iSLEEPS. A different and harder question is what happens when a model built for healthy sleep is applied to this cohort as published. The three architectures are our own: a four-channel CNN, the same CNN with a BiLSTM over epochs, and StagingSeqNet, a deeper convolutional-recurrent stager we built for this comparison. Naming them matters only in that none is a weak straw man; each reaches 0.81 to 0.87 on the healthy corpus it was trained on. We trained them on the Sleep-EDF Cassette subset and applied each to all 99 iSLEEPS patients zero-shot, with no adaptation of any kind. This is the middle block of Table 4, set against the same architectures retrained on this cohort.

The result is not a degradation but a collapse. All three stage healthy sleep competently, 0.806 to 0.868 accuracy and κ 0.73 to 0.83; on stroke the same checkpoints reach 0.307 to 0.325 with κ 0.09 to 0.11, which is close to chance for a five-class problem with this prior. The failure mode is more informative than the headline: each model labels 66–78% of stroke epochs Wake against a true 26.6%, and N2 recall falls to 0.032 on the stage that is 42% of the data. These models do not lose accuracy gracefully; they stop recognising sleep architecture at all.

The control is what makes this interpretable. A healthy-trained stager scoring 0.31 on iSLEEPS is evidence of a cohort effect only if the same checkpoint, through the same preprocessing, scores properly on healthy sleep; otherwise a broken input pipeline and a genuine domain gap produce identical numbers. Run through our pipeline on held-out Sleep-EDF recordings, the strongest checkpoint holds at 0.8682 (0.8667 ± 0.0415 per recording), so the pipeline is correct and the collapse belongs to the cohort.

Two bounds, reported together. Sleep-EDF is recorded on Fpz-Cz and Pz-Oz where iSLEEPS uses C4:M1 and O2:M1, so part of this gap is montage shift, not pathology, and quoting the zero-shot number alone would overstate the case. Deep models trained *on* iSLEEPS with the matched montage reach 0.61–0.69 (upper block), which puts the cost of pathology alone at roughly 0.19 against healthy-sleep performance; healthy-trained and zero-shot, with the montage shift included, the cost is 0.56. The claim is bounded by those two figures together.

The same argument run in reverse. Every figure above concerns other architectures, which leaves the obvious question unanswered: whether 0.739 is low because the cohort is hard or because the proposed model is weak. So we cross-validated it on healthy sleep, ten-fold and subject-independent over the 21 Sleep-EDF subjects, using the neural branch alone since that corpus has no cardiorespiratory channels. It reaches 0.856 ± 0.027 accuracy with κ 0.800 (final row of Table 4). That places it among the healthy-sleep stagers in the middle block, against 0.741 for the identical architecture on stroke. The 0.115 difference is what the cohort costs a model that is demonstrably capable of more, which is a stronger statement than any comparison against other people's architectures can make.

The cohort does not cost every architecture equally. Because the three reimplemented models carry a healthy-sleep score produced by the same code under the same protocol, the table supports a comparison it could not before: not merely where each model lands, but how much it loses in the crossing. TinySleepNet falls 0.148 from healthy sleep to stroke ($0.824 \rightarrow 0.676$), SleepTransformer 0.140 ($0.823 \rightarrow 0.683$) and U-Time 0.145 ($0.791 \rightarrow 0.646$), against 0.115 for the proposed model ($0.856 \rightarrow 0.741$). It is therefore not simply higher at both ends; it degrades least. We report this as an observation. Four paired differences will not support a significance test, and the comparison is between systems rather than a controlled ablation, since the proposed rows read engineered features and a frozen embedding over four derivations while the three reimplemented baselines read a single raw channel. The direction is nonetheless the one the rest of the paper predicts: a representation that encodes sleep physiology explicitly survives a disordered cortex better than one a network must discover for itself from fewer than a hundred patients.

Folds are split by subject, not by recording, because Sleep-EDF Cassette records each subject on two nights; splitting on filenames would place the same person on both sides of a fold.

N. STREAM REPRESENTATIONS

The architecture makes two representation choices (engineered features plus a frozen self-supervised embedding on the neural side, a learned convolutional encoder over the raw signal on the cardiorespiratory side), and both were selected by experiment, not assumed. Each family below holds every other hyperparameter fixed and varies one thing, over the same three seeds and ten patient-independent folds, with Holm correction within the family and every test on the ten fold-means.

The cardiorespiratory features are the bottleneck, and part of the gap needs no new information. Each block of Table 10 holds the other stream at its baseline, so no single row there is the full model. The upper block replaces the 14 engineered cardiorespiratory features with progressively richer representations of the same underlying signal. Every step is significant, and the ordering is instructive. A *random* nonlinear expansion of the same 14 numbers recovers 0.018 AUC: that gain needs no new measurement at all, only more capacity to read what was already computed, which says the summary features leave nonlinear structure unextracted. A random linear projection of the raw waveform adds a further 0.016, so part of what the features discard is recoverable without learning anything. The learned convolutional encoder reaches 0.782, 0.084 above the engineered features.

What this establishes is that the representation of the cardiorespiratory stream, and not the detector on top of it, is what limited the respiratory read-out. We therefore claim the change of representation itself, and make no claim for the encoder that implements it.

TABLE 10. What each stream should be represented as. One factor varies per block, everything else held fixed; cells are mean ± SD over thirty fold-values. ´ and Holm *p* are against each block's engineered-feature baseline, the latter on the ten fold-means. "adopted" marks the choice carried into the final model.

Representation	Variant	Resp. AUC ↑	Staging acc ↑	Δ	Holm p
Cardiorespiratory	14 engineered features	0.698 ± 0.051	0.736 ± 0.014	—	—
	random nonlinear expansion of the 14	0.715 ± 0.047	0.737 ± 0.018	$+0.018$	0.012
	random linear projection of raw signal	0.731 ± 0.044	0.735 ± 0.021	$+0.034$	0.012
	learned CNN, 0.155 M (adopted)	$\mathbf{0.782 \pm 0.040}$	0.740 ± 0.019	$+0.084$	$\mathbf{0.006}$
Neural	188 engineered features	0.685 ± 0.041	0.741 ± 0.016	—	—
	frozen pretrained encoder alone	0.681 ± 0.033	0.712 ± 0.033	-0.029	0.008
	features + pretrained (adopted)	0.690 ± 0.037	$\mathbf{0.749 \pm 0.018}$	$+0.008$	$\mathbf{0.006}$
	features + two pretrained encoders	0.684 ± 0.043	0.745 ± 0.018	$+0.004$	0.23

On the neural side a pretrained encoder does not replace expert features, but does add to them. The lower block of Table 10 varies the neural representation with the cardiorespiratory stream held at the engineered features. Used alone, the frozen encoder is significantly *worse* than the 188 features (-0.029, Holm $p = 0.008$): pretraining on healthy scalp EEG does not by itself substitute for physiology chosen for this task. Concatenated with the features it adds $+0.008$ accuracy (Holm $p = 0.006$), and a second pretrained encoder on top of the first adds nothing further (-0.004, $p = 0.56$).

The reading we take is narrow, because it is what the numbers support: expert physiology and self-supervised pretraining encode partly different information, so their union beats either alone; but the complementarity saturates after one pretrained encoder, so this is not a general argument for stacking foundation models.

O. CALIBRATION AND SELECTIVE PREDICTION

A clinical reader needs to know not only how often the model is right but whether it knows when it is likely to be wrong. Two measurements answer that, both computed on held-out epochs with the raw posteriors before HMM smoothing.

The model is *overconfident*: expected calibration error is 0.076, and the gap runs one way in every confidence bin: epochs it scores above 0.9 are correct 88% of the time. Stated confidences should therefore be read as ordered, not as probabilities.

Ranking, however, is what abstention needs. Applying split-conformal prediction with the fold's validation patients as the calibration set gives empirical coverage that tracks the target closely (0.894 at a 0.90 target, 0.943 at 0.95). At $\alpha = 0.10$ the model returns a single stage for 45.8% of epochs and is correct on 86.9% of those, against 59.1% on the epochs where it returns a larger set. The abstention is therefore selecting the genuinely hard epochs, and it concentrates where a scorer would expect: N1 receives a singleton set only 19% of the time, the lowest of any stage and the one human scorers agree on least.

P. EXTERNAL VALIDATION ON TWO INDEPENDENT CORPORA

Ten-fold cross-validation is patient-independent but remains within one corpus, one centre and one acquisition protocol. We therefore trained a single model on all 99 iSLEEPS patients and evaluated it, unchanged, on two public corpora with no dataset-specific tuning: no fine-tuning, no threshold search, no re-fitted model or decision threshold (per-recording feature standardisation is applied, as in training), and hidden-Markov transitions estimated from iSLEEPS alone (Table 11).

The two corpora test different things. **Sleep-EDF** [41] is healthy sleep recorded on a frontal montage (Fpz-Cz, Pz-Oz) with no cardiorespiratory channels, so it tests the staging head across both a cohort shift and a montage shift. **ISRUC-Sleep** [52] is a sleep-disorders cohort recorded on the same central and occipital derivations we use (C4, C3, O2, O1 against the contralateral mastoid), with SpO_2, airflow, effort and ECG, and with obstructive, central and mixed apneas and hypopneas scored, so it tests the staging head *without* a montage confound and provides the only external test of the respiratory head. Because Sleep-EDF carries no cardiorespiratory channels, that branch is fed zeros there. Zeroing an input the network trained on is a train/test mismatch, not an ablation, so we also trained a model with the branch zeroed throughout: it reaches 0.774 accuracy against the 0.794 reported. We read the pair as bracketing the true transfer figure, not as confirming the higher one. The mismatched run could be gaining from batch-normalisation statistics that never saw zeros, in which case 0.774 is the more defensible number, and a reader who prefers it loses nothing in our argument: both sit above the in-domain 0.739. Both corpora also differ from iSLEEPS in montage, so both transfer figures are lower bounds and not clean estimates. ISRUC records each subject twice; respiratory-event prevalence is 10.1% on the first night and 2.4% on the second, so we report the two separately instead of pooling a difference that has nothing to do with the model.

The respiratory read-out transfers. On ISRUC night one the model reaches 0.780 AUC, and it does so handicapped, because ISRUC provides no pulse channel and no separate effort trace, leaving two of the seven cardiorespiratory inputs zero-filled. A detector that had latched onto one centre's equipment would not survive that; this one does. The comparison needs its estimators matched to mean anything, and we state them instead of placing the two numbers side by side. The external figure is pooled over all epochs of the night, because per-recording AUC is unusable on this corpus: event prevalence ranges from 0% to 22% across the sixteen recordings, and a night with almost no events returns a near-chance value whatever the model does. The in-domain headline 0.782 is a ten-fold mean. Pooled in-domain, the like-for-like quantity, is 0.778. The external result therefore sits within 0.003 of the in-domain one on either estimator, and nothing in the claim rests on which is chosen. Night two falls to 0.723, tracking its much lower event prevalence (2.4% against 10.1%).

Staging moves in opposite directions on the two corpora, for reasons the data identify. On Sleep-EDF the model *gains* 0.084 of κ over its in-domain score. A representation that had memorised corpus-specific artefacts would degrade out of corpus, so the straightforward reading is that the features are physiological and that the ceiling reported throughout this paper is a property of *stroke sleep* rather than of the model. The learning curve reaches the same conclusion (Section VI-L) by a different route. On ISRUC staging falls by 0.090 accuracy, and the class mix explains much of it: ISRUC carries 15.6% N1 against iSLEEPS' 10.2% and Sleep-EDF's 6.6%, and N1 is the weakest stage in every cohort we have measured (recall 0.168 on ISRUC, 0.294 on Sleep-EDF, and the lowest per-class F1 in-domain).

A montage explanation the data now rule out. An earlier version of this work observed N3 recall of 0.554 on Sleep-EDF with 44% of N3 epochs read as N2, and attributed it to slow-wave activity being frontally dominant while the model reads central and occipital derivations. That account made a prediction: N3 should recover on ISRUC, which uses our derivations exactly. Under the final model the prediction fails, and it fails in the direction that rules the explanation out. N3 recall is 0.841 on Sleep-EDF, the mismatched frontal montage, with 12.7% of N3 read as N2, and 0.630 on ISRUC, the matching one, with 34.8%. Electrode position is therefore not what governs N3 transfer here. The only component that changed between the two models is the neural representation (Section VI-N), which points instead at how the cortex is encoded, though with one external pass per corpus and eight ISRUC subjects we can identify the effect without attributing it confidently. The prediction stays on the record, and it was wrong.

Two limits keep this in proportion. Accuracy across cohorts with different class balance is not directly comparable, which is why κ is reported alongside. And ISRUC contributes only eight subjects. Per-recording accuracy spans 0.458–0.748, so the ISRUC figures in Table 11 are quoted to three decimals for consistency with the rest of the paper and should be read to two; per-recording respiratory AUC spans 0.515–0.848, the latter tracking event prevalence closely, so the low-prevalence nights give near-chance estimates.

Q. MODEL COMPLEXITY AND CAPABILITY

Table 12 sets the proposed model beside representative alternatives on size and capability. At 2.76 M trainable parameters the model is an order of magnitude smaller than DeepSleepNet (∼21 M) and the corpus's own CNN-ResNet18 (∼11 M), and comparable to SleepTransformer, so size is not where it wins. It delivers both a competitive stage and a respiratory read-out from one pass over the recording, and none of the staging-only models, at any size, can supply the second output.

TABLE 11. External validation, zero-shot: one model trained on all 99 iSLEEPS patients, no dataset-specific tuning. In-domain figures are ten-fold means. Sleep-EDF has no cardiorespiratory channels, so the respiratory head is unscored there; the two ISRUC nights differ in event prevalence and are reported separately.

Corpus	Cohort	Rec.	Acc ↑	κ ↑	Resp. AUC ↑
iSLEEPS (in-domain)	stroke	99	0.739	0.634	0.782
Sleep-EDF	healthy	40	**0.794**	**0.718**	—
ISRUC, night 1	sleep-disord.	8	0.649	0.536	**0.780**
ISRUC, night 2	sleep-disord.	8	0.575	0.426	0.723

TABLE 12. Model size and clinical capability. The proposed model is mid-sized and is the only one that produces both a stage and a respiratory-event label.

Model	Input	Params ↓	Outputs
CNN-ResNet18 [17]	1 EEG	∼11 M	stage
DeepSleepNet [2]	1 EEG	∼21 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	7 ch feat + raw	**2.76 M**	**stage + apnea**

VII. DISCUSSION

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. The learned per-epoch embedding separates cleanly by sleep stage while the respiratory structure stays diffuse, mirroring the strong staging and the modest respiratory AUC.

Staging accuracy on this cohort saturates below 0.75 for every model family we evaluated, and the proposed model reaches 0.739 while adding a respiratory capability the staging models lack. The respiratory read-out does not depend on the labeling channel: it survives deletion of the airflow channel and rests on respiratory effort, desaturation, and arousal, the same signs a physician reads at the bedside.

The 0.739 staging figure should be judged on this cohort, where healthy-sleep accuracy does not transfer. Deep architectures developed on healthy sleepers collapse here: DeepSleepNet reaches 0.615, CNN-ResNet18 0.617, and a network pretrained on the healthy Sleep-EDF corpus and applied to this cohort 0.623, each roughly fifteen to twenty points below its healthy accuracy of about 0.85. The injured brain distorts the very micro-events, spindles, slow waves, and atonia, that those models learned to read. What holds up is physiologically grounded modeling on engineered features, at 0.72 to 0.75, which matches the published baseline on this corpus. The lesson is that healthy-sleep deep learning fails on the injured brain, and this cohort requires cohort-specific, physiologically grounded, multimodal modeling.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is the expected outcome: the second modality adds a new capability while leaving staging unchanged. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

A. LIMITATIONS

The system is offline, and mildly transductive. Two properties follow from the design and neither is visible in the numbers. Staging uses whole-night Viterbi decoding, so no epoch can be scored until the recording ends, which rules out the real-time use a bedside monitor would need. And features are standardised per patient, which at test time uses that night's own statistics; the model never sees another patient's labels, so this is not label leakage, but it is a transductive step and a strictly causal deployment would need running or population statistics instead. We have measured what that substitution costs, and within the resolution of these folds it costs nothing. Repeating the ten folds with the standardisation statistics pooled over each fold's training patients and applied unchanged to the held-out ones moves accuracy by $+0.003$, κ by $+0.004$, and respiratory AUC by -0.000, none of which is distinguishable from zero when paired across folds (accuracy $p = 0.28$, AUC $p = 1.00$). The per-recording statistics are a convenience rather than a crutch, and the transductive step is not what the numbers rest on. The whole-night decoding is the real obstacle to bedside use, and it is untouched by this.

A prediction of ours that failed. The montage account of N3 transfer described in Section VI-P does not survive retesting under the final model, and we report it as falsified instead of removing it. Two cautions follow for every other per-class figure in that section: each rests on a single external pass with no fold structure, so none carries an interval, and ISRUC contributes eight subjects, at which size a per-class rate is an estimate with a width we cannot quote. Per-class transfer should be read as direction, not magnitude.

The evidence comes from a single center and 99 patients. Section VI-P shows both heads transfer to independent corpora without tuning, but neither external corpus is a stroke cohort, so every stroke-specific claim rests on iSLEEPS alone; multi-center validation on a second stroke cohort is required before clinical use, and no such public corpus currently exists. The external respiratory result rests on eight subjects, two of whose seven cardiorespiratory inputs were unavailable, so it establishes that the read-out transfers, not how well. Respiratory detection, at 0.782 AUC, is a useful signal but not yet a screening-grade one, and it is scored per epoch. At event level it detects 55% of events at 2.6 false alarms per hour and undercounts the event rate by more than half, so it cannot substitute for a scored index. The staging head is also overconfident (expected calibration error 0.076), so its posteriors should be used for ranking and abstention. Staging sits at the cohort ceiling, so the model does not raise staging accuracy above the published baseline for this corpus. Finally, the model is feature-based: it inherits the assumptions of the engineered representations and is only partly an end-to-end raw-signal learner, since the neural stream reads engineered descriptors and a frozen embedding while the cardiorespiratory stream is trained on the raw signal, a deliberate trade for data efficiency on a small cohort.

B. FUTURE WORK

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

VIII. CONCLUSION

We built a model that reads the neural and cardiorespiratory channels of a stroke patient's polysomnogram and, from one pass, stages their sleep and detects their respiratory events. Deep networks built for healthy sleep collapse to 0.61–0.72 accuracy on this cohort, so its 0.739 staging accuracy is at the cohort ceiling for every model family. The contribution is the second output, a respiratory read-out at 0.782 AUC that survives removal of the airflow channel, showing it reads the underlying physiology. Reading the whole recording lets one model report on breathing as well as on sleep, in the population where breathing matters most. Code and features are released.

DATA AND CODE AVAILABILITY

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [14]. Our code, engineered features, and reproduction notebook are released at https://github.com/wshuv-o/isleeps-sleep-staging.

ETHICS

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [14]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

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APPENDIX A THE 188 NEURAL DESCRIPTORS

The neural feature vector is 23 descriptors computed on each of the seven neural channels (C4:M1, C3:M2, O2:M1, O1:M2, E1:M2, E2:M2, chin EMG), giving 161, plus 27

TABLE 13. Recent single-channel sleep-staging methods: the accuracy each reports on healthy or general cohorts, set against its accuracy on the subacute-stroke cohort (iSLEEPS) for the five architectures we evaluated on it. A dash marks a value we did not evaluate.

Method	Year	Approach	Reported acc (dataset)	On iSLEEPS
DeepSleepNet [2]	2017	Two representation-learning CNNs followed by a bidirectional LSTM sequence model, single-channel EEG.	0.82 (Sleep-EDF)	0.615
U-Time [7]	2019	Fully convolutional encoder–decoder segmenting a whole night at once, no recurrence over epochs.	≈0.79 (Sleep-EDF)	0.646
TinySleepNet [4]	2020	Compact convolutional stack with a single unidirectional LSTM, single-channel EEG.	0.854 (Sleep-EDF)	0.676
AttnSleep [5]	2021	Multi-resolution CNN with multi-head self-attention over epochs, single-channel EEG.	0.844 (Sleep-EDF)	0.716
SleepTransformer [18]	2022	Transformer encoder with epoch-level interpretability and uncertainty estimates.	0.814 (Sleep-EDF)	0.683
1D-ResNet-SE-LSTM [49]	2023	Residual network with squeeze-and-excitation blocks and an LSTM on raw EEG.	0.864 (Sleep-EDF)	0.708
Micro SleepNet [50]	2023	Lightweight convolutional model for real-time mobile staging.	0.828 (Sleep-EDF-20)	0.647
AT-BiLSTM [51]	2021	Attention-augmented bidirectional LSTM, single-channel EEG.	0.838 (Sleep-EDF)	0.692
MM-Net (proposed)	–	Two feature streams (neural and cardiorespiratory) fused into a BiLSTM that jointly stages sleep and detects respiratory events from the fourteen-channel polysomnogram.	–	**0.739**

event descriptors that apply only to the channels whose physiology defines them, giving 188 in total. Every epoch is 30 s at 100 Hz, so 3000 samples, and every descriptor is computed on that epoch alone with no smoothing across epochs; the temporal context is supplied later by the recurrent decoder rather than by the features.

Per-channel descriptors ($23 \times 7 = 161$)

The power spectrum is Welch's estimate with a 256-sample segment and 128-sample overlap. Writing $P(f)$ for that spectrum and $T = \sum_f P(f)$:

- *Band powers, ten.* Absolute and relative power in delta (0.5–4), theta (4–8), alpha (8–13), sigma (11–16) and beta (16–30) Hz. Alpha and sigma overlap deliberately: sigma is the spindle band, not a subdivision of alpha.
- *Spectral shape, four.* $\log T$; spectral entropy $-\sum_f \tilde{P}(f)\log\tilde{P}(f)$ on the normalised spectrum $\tilde{P} = P/T$; the 95% spectral edge frequency; and the mean frequency $\sum_f fP(f)/T$.
- *Time domain, six.* Standard deviation, peak-to-peak, root mean square, skewness, kurtosis, and zero-crossing rate.
- *Hjorth, three.* Activity σ_x^2, mobility $\sigma_{x'}/\sigma_x$, and complexity $(\sigma_{x''}/\sigma_{x'})/\text{mobility}$, from the first and second differences [47].

Event descriptors (27)

These encode the graphoelements a scorer reads, and each is applied only where it is physiologically defined. All band-pass filters are fourth-order Butterworth, applied forward and backward so the phase is unchanged.

- *Spindles, three per EEG channel (12).* The channel is filtered to 11–16 Hz and its analytic envelope taken by Hilbert transform. The detection threshold is the 90th percentile of that envelope *within the epoch*, so it adapts to the epoch's own amplitude rather than to a fixed microvolt level. Density is the number of upward crossings of that threshold; the other two are the envelope's mean and standard deviation.
- *Slow waves, two per EEG channel (8).* The channel is filtered to 0.5–4 Hz; we take the peak-to-peak amplitude and the mean absolute amplitude. No duration or morphology criterion is imposed, so these are amplitude summaries of slow-wave activity rather than discrete detections.
- *Ocular, two per EOG channel (4).* The log mean squared first difference, and the 95th percentile of its absolute value, which together separate the fast deflections of REM from slow drift.
- *Chin EMG, three.* Log root-mean-square, the 90th percentile of the absolute signal, and the log variance of the first difference, which track tonic level and its loss in REM atonia.

Artifact handling

No epoch or channel is rejected, and no amplitude threshold is applied. A corrupted channel therefore reaches the model, which is deliberate: the ablations in Section VI-F show the streams are redundant enough to tolerate one, and a rejection rule would have to be tuned on this cohort and would then be one more thing that does not transfer. The per-patient standardisation below does the work that a gain-matching step would otherwise do. Non-finite values, which arise only from a flat channel, are replaced by zero.

Aggregation and scaling

There is no aggregation across epochs. Each descriptor is standardised per recording, subtracting that recording's mean and dividing by its standard deviation, which is the transductive step discussed in Section VII-A.

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